

TED UNIVERSITY

Department of Computer Engineering

CMPE 491 – Senior Design Project I

PROJECT SPECIFICATIONS

Improving Object Detection Performance in SAR Imagery Using Super-Resolution Techniques

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Improving Object Detection Performance in SAR Imagery Using Super-Resolution Techniques

Project Website
https://ilkeozcendek.github.io/keskin/

Related Program
SAYZEK-ATP 2026–2027 · "Süper Çözünürlük Teknikleri ile SAR Görüntülerinde Tespit Başarımının Artırılması"
[under evaluation]

1. Introduction

1.1 Project Description and Scope

Synthetic Aperture Radar (SAR) imaging ensures all-weather, day-and-night observation, which is essential for critical monitoring and disaster response. However, compared to optical images, SAR data is impacted by speckle noise and low spatial resolution backscattering, which blur fine spatial details and blur the boundaries of small or closely packed targets. These limitations restrict object detection in SAR images.

These resolution constraints severely limit the performance of modern deep learning object detection architectures, causing high alarm rates and the missed detection for the targets. However, purchasing high-resolution SAR data directly is often cost-prohibitive or constrained by satellite revisit times. Deep learning, which is based on super-resolution, allows a computational approach to enhance image clarity and retrieve high-frequency details from datasets.

Research Question

How does deep learning based super resolution pre-processing improve object detection performance on SAR imagery?



Figure 1. *Planned processing pipeline.*

In Scope

S1Object detection on annotated SAR dataset.

S2Synthetic low-resolution inputs to be created from the original high-resolution images.

S3Two learned super-resolution models (candidates: EDSR and SRGAN) compared with upsampling, building on SAR super-resolution work .

S4Detector usage such as (CNN Detector (Convolutional Neural Network) or YOLOv8) will be evaluated.

If Time Permits

E1 Exploring advanced loss functions

E2 Generalizability across varying SAR incidence angles for dataset

Out of Scope

X1 On-board processing.

X2 Target type classification.

1.2 Constraints

Constraints governing our project in alignment with academic standards.

Constraint	Effect on the Project	Planned Response
Data and annotations	SAR target data availability and annotation quality.	Utilizing ESA SNA for consistent pre-processing.
Computing resources	Limited GPU memory and training speed.	Optimized Batching
Time	Course deadlines.	Follow a strict internal task schedule.
Economic	GPU and costing software of SAR.	Using more reachable source tools and academic resources.
Environmental	Energy consumption	Avoid redundant runs and use early stopping.
Social	Benefits for disaster management and safety.	Improve detection accuracy.

Constraint	Effect on the Project	Planned Response
Political	Dual-use risks of radar monitoring.	Strictly follow open academic guidelines.
Ethical	Licensing, honest reporting synthetic details (Section 1.3).	Compare outputs.
Health and safety	No direct risk; consequences of missed or false detections if used operationally.	Present results strictly.
Manufacturability	Code portability and deployment ease.	Provide a modular, reproducible software repository such as (Github or Gitlab).
Sustainability	Long-term code and data usability.	Document progress clearly on the project.

Table 1. *Project constraints.*

1.3 Professional and Ethical Issues

Through open data utilization, truthful reporting, and responsible civilian application, this project conforms to the ACM [5], IEEE [6], and Software Engineering [7] codes of ethics.

Data Licensing and Attribution: Every dataset, library, and trained models will be utilized completely in accordance with the terms of each license.

Honest Reporting There will be complete transparency in the reporting of our experimental results. In order to avoid making any unjustified claims, this involves situations in which super-resolution augmentation produces insignificant results.

Synthetic Detail Generative super-resolution models (such as SRGAN) can introduce artifacts or structures absent in the original scene.Super-resolved images will not be displayed as unprocessed ; instead, they will be clearly labeled.

Dual Use Target detection can be used for monitoring. However, this study uses only public data for disaster response , clearly stating all limits.

Personal Data Public datasets contain no personal data, and team members' personal details will not be published on the project website.

1.4 Definitions and Abbreviations

Table 2. *Definitions and abbreviations.*

Term	Definition
SAR	Synthetic Aperture Radar.
Speckle	Granular noise in SAR images caused by the coherent nature of radar signals.
SR	Super-resolution: reconstructing a higher-resolution image from a lower-resolution input.
HR / LR	High resolution / low resolution.
Scale factor	The ratio between HR and LR image dimensions.
CNN	Convolutional Neural Network.
GAN	Generative Adversarial Network: a generator and a discriminator trained against each other.
EDSR	Enhanced Deep Super-Resolution network.
SRGAN	Super-Resolution Generative Adversarial Network.
Bicubic interpolation	A non-learning upsampling method that estimates new pixels from the surrounding 4 × 4 pixels.
YOLO	You Only Look Once, a single-stage object detector.
IoU	Intersection over Union between a predicted and a ground-truth box.
Precision/ Recall	Share of predicted ships that are correct / share of real ships that are found.
F1 score	Harmonic means of precision and recall.
AP / mAP	Average precision / mean average precision.
PSNR	Peak signal-to-noise ratio.

Term	Definition
SSIM	Structural similarity index measuring.
COCO	Common Objects in Context; a widely used object-detection benchmark whose annotation format and object-size thresholds are used in this project.
Small / medium / large objects	COCO size groups meaning box area pixels.
HRSID / SSDD	High Resolution SAR Images Dataset.

2. Requirements

The requirements below are written as “shall” statements. Each requirement has an ID, a source (objectives O1–O3, outcomes D1–D3, topic or a team decision) and verification method.

2.1 Data Requirements

HRSID [1] and SSDD [4] are the current candidates. The requirements below apply to when the dataset is selected.

Table 3. *Data requirements.*

ID	Requirement	Source	Verification
DATA 01	For every High Resolution image used, the counterparts shall be generated with the same blur and downsample pipeline.	O1	Pairing and imaging size checking
DATA 02	All derivatives of image shall stay in the same data split.	Team	Split list check
DATA 03	Bounding boxes shall be rescaling with each image that every condition shall be evaluated on the same set of test targets, with object size groups defined from the High resolution reference boxes.	O1	Annotation and test set check
DATA 04	Train, validation and test splits shall be fixed before any training.	Team	Versioned split files

ID	Requirement	Source	Verification
DATA 05	Dataset name, version, licences and download date shall be recorded.	Team	Data card in the repository

2.2 Functional Requirements

Table 4. *Functional requirements.*

ID	Requirement	Source	Verification
FR 01	The system shall prepare low resolution(LR) and high resolution (HR) pairs from the selected dataset.	O1,	Generated pairs
FR 02	The system shall train two SR architectures on the Synthetic Aperture Radar training will be evaluate them	O2, D1,	Saved models and outputs
FR 03	The system shall produce bicubic images as a baseline.	O2	Baseline outputs
FR 04	The system shall train and evaluate detectors on each input.	O3,	Detector runs per condition
FR 05	The system shall compute image quality and detection metrics and export them.	D3	Result tables
FR 06	The system shall apply each trained SR model to (LR) low resolution images of all data splits, so that the detector can be trained and tested on SR outputs.	O3, P1	SR image sets for each split

2.3 Nonfunctional Requirements

Table 5. *Nonfunctional requirements.*

ID	Requirement	Source	Verification
NFR 01	Every experiment shall be reproducible from configuration files and the recorded versions.	D2	Rechecking
NFR 02	Every run shall be logged with configuration and metrics.	D2	Run logs

ID	Requirement	Source	Verification
NFR 03	Training shall run on the hardware available to the team (GPU servers or a cloud service such as Google Colab), and the hardware used shall be recorded.	Team	Hardware record
NFR 04	Training and inference time and memory use shall be recorded.	Team	Timing table
NFR 05	The repository shall include step by-step instructions to reproduce the results.	D2	README review

2.4 Experimental and Evaluation Requirements

We compare the five input conditions in Table 6 using two detector protocols. Four of these conditions use low-resolution images, which we prepare at both ×2 and ×4 scales from the same HR images. For each experiment, we keep the detector architecture, hyperparameters, training epochs and data splits the same. What changes is the type of image given to the detector and its input size, following EVAL-05. This way, we can compare the detection results without changing the other experimental settings.

Table 6. *Input conditions.*

Input Condition	Role in the Comparison
Low resolution input	Baseline without super resolution
Bicubic upsampling	Non learning upsampling baseline
SR model A	Learned SR aimed at faithful reconstruction (candidate: EDSR)
SR model B	Learned SR aimed at perceptual sharpness (candidate: SRGAN)
High-resolution reference	Reference based on the original images which are always available because the LR inputs are created from them

Evaluation Protocols

P1 **Retraining:** In this protocol, we train and test the detector separately for each input condition at ×2 and ×4 scales. We use the same hyperparameters, training epochs and data splits throughout the experiments. However, the images used for training and their input sizes depend on the condition, as described in EVAL-05. We want to find out whether

training the detector with enhanced images makes a difference in detection accuracy. This is also one of the main objectives of our SAYZEK project.

P2 **Fixed detector:** We train one detector on the HR training images and use it to test all other input conditions without training it again. Following EVAL-05, LR images are tested at their original size, which means objects appear smaller than they do in the HR images used for training. Bicubic and SR images are tested at the HR size. By comparing the results, we can see how the detector performs on lower-resolution images and whether bicubic upsampling or super-resolution helps improve its performance.

Table 7. *Evaluation requirements.*

ID	Requirement	Source	Verification
EVAL-01	PSNR and SSIM must be reported for all bicubic and SR outputs at both ×2 and ×4 scales using the HR images as a reference	O2, D3	Image - quality table
EVAL-02	For each input condition and scale, precision and recall must be reported using the confidence threshold that gives the highest F1 score on the validation set. We also report mAP at IoU 0.5 for both P1 and P2. Under P1, the confidence threshold is selected separately using the validation set of each condition. Under P2, the threshold is selected once using the HR validation set.	O3, D3	Detection tables
EVAL-03	Detection results must also be reported separately for small, medium and large objects. Object sizes are determined using the COCO area thresholds based on the HR reference bounding boxes [1].	D3	Per size table
EVAL-04	All input conditions must be evaluated using the same held out test images and ground truth annotations.	Team	Split and annotation check

ID	Requirement	Source	Verification
EVAL-05	Each input condition is tested at its corresponding image size without resizing. LR images are used at half or one quarter of the HR dimensions, while bicubic, SR, and HR images are tested at the HR size. If the width or height is not a multiple of the detector's stride, padding is added instead of resizing. The input dimensions and any padding applied are recorded for each run.	Team	Run configuration record
EVAL-06	PSNR and SSIM must be calculated using single-channel images at the HR size. The same data range must be used for all outputs, and a border of s pixels must be excluded for each scale factor s.	Team	Metric script review

2.5 Acceptance Criteria and Traceability

We do not expect super-resolution to always improve detection results. Finding out whether it helps is the reason we are carrying out these experiments. For this reason, we will not judge the success of the project by a specific accuracy score. Instead, we need to finish the planned comparisons, reproduce the results and include them in our report. Table 8 explains what is required for each objective and expected outcome and how we will check their completion.

Table 8. *Traceability from objectives to requirements.*

Objective / Outcome	Covered by	Accepted When
O1 · LR–HR dataset	DATA-01–05, FR-01	For every selected HR image we have both ×2 and ×4 image pairs. All of the pairs must pass the image size and annotation checks and the data splits remain fixed
O2 · Two SR models	FR-02, FR-03, EVAL-01, EVAL-06	Test both SR models and bicubic upsampling at ×2 and ×4 scales. For each model and scale, we report the PSNR and SSIM results.
O3 · SR in detection	FR-04, EVAL-02–05	Complete both P1 and P2 for all five input conditions, using ×2 and ×4 scales where applicable

Objective / Outcome	Covered by	Accepted When
D1 · Trained SR models	FR-02, EVAL-01, EVAL-06	Save both trained SR models along with the settings used during training. This allows us to generate the same outputs again when needed.
D2 · Reproducible pipeline	NFR-01, NFR-02, NFR-05	Run the experiments again using the same instructions, random seeds and configurations. We then check the new results against the original tables. If there are any differences, we record them alongside the reported values.
D3 · Gain analysis	FR-05, EVAL-01–06	Gains and losses against the LR and bicubic baselines are reported per condition, scale and size group, including cases where SR does not help
SAYZEK topic	FR-01, FR-02, FR-04, P1, EVAL-02	A detector retrained on super-resolved images (P1) is evaluated and its accuracy change against the LR and bicubic baselines is reported

3. Literature Review

The three studies below are reviewed with respect to the requirements in Section 2: [1] informs the data and size-group requirements, [2] the SR requirements, and [3] the evaluation protocol.

3.1 Object Detection in SAR Imagery

[1] *S. Wei et al.* · *IEEE Access, 2020*

Introducing the High-Resolution SAR Images Dataset (HRSID) , Wei et al. demonstrated that "spatial resolution plays a crucial role" in SAR target detection. While establishing the importance of high-resolution data, the study leaves computational super-resolution unexamined, directly supporting our dataset selection.

3.2 Super-Resolution for SAR Images

[2] *P. Addabbo et al.* · *IEEE Access, 2023*

Addabbo et al. developed a CNN model called DC2SCN to improve the resolution of SAR images. The model processes the real and imaginary parts of complex SAR data separately, so that both amplitude and phase information are preserved. The researchers created low- and high-resolution image pairs by processing COSMO-SkyMed StripMap data at different bandwidths, which gave about 20,000 image segments of 512 × 512 pixels. On real data, the model increased PSNR by about 14 dB compared to the degraded input. This result supports FR-02, which requires our SR models to be trained on SAR images instead of using weights learned only from optical photos. However, the study only measured image quality and did not test whether the improved images help object detection. In addition, KESKİN will work with amplitude images and create LR images by blurring and downsampling (DATA-01), which is simpler than the bandwidth-based method used in [2].

3.3 Effect of Super-Resolution on Detection

[3] *J. Shermeyer and A. Van Etten* · *CVPR Workshops (EarthVision), 2019*

Shermeyer and Van Etten [3] studied how super-resolution affects object detection in optical satellite images. They tested VDSR and their Random Forest Super-Resolution (RFSR) model at 2×, 4× and 8× scales on images with resolutions between 30 cm and 4.8 m. For object detection, they used the SIMRDWN framework, which includes YOLO and SSD models. The detectors were trained on both original and super-resolved images. Without super-resolution, mAP dropped from 0.53 at 30 cm to 0.11 at 4.8 m. The largest improvement came from increasing the resolution from 30 cm to 15 cm, where mAP improved by 13–36%. At lower resolutions, the improvements were much smaller. They also looked at how object size in pixels affected detection performance.

The study also provides a basis for our approach, where we retrain the detector under different input conditions (P1), measure detection performance with mAP (EVAL-02), and analyse the results based on object size (EVAL-03). However, their experiments focused on optical satellite images rather than SAR images. Since SAR images have different scattering characteristics and contain speckles, we cannot assume that super-resolution will provide the same improvements. Our project aims to investigate this difference.

3.4 Research Gap and Implications for KESKİN

The studies reviewed here leave a clear gap. HRSID [1] provides a high-resolution benchmark for object detection in SAR imagery but does not study super-resolution. Addabbo et al. [2] used a CNN to improve the resolution of SAR images. Their study, however, only looked

at image quality and did not evaluate detection performance.Shermeyer and Van Etten [3] show that super-resolution can improve detection, but only in optical imagery. KESKİN addresses this gap by testing, under one controlled protocol, whether super-resolving SAR images improves object detection. We compare two SR models with bicubic upsampling at ×2 and ×4 scales. We then look at the image quality results and compare detection performance for small, medium and large objects.

Implications for the Requirements

L1 [1] → **DATA-03, EVAL-03:** In HRSID, more than half of the annotated objects are considered small according to the COCO area thresholds. We keep the test targets the same for every input condition and use the HR reference bounding boxes to determine the size of each object.

L2 [2] → FR-02: A CNN trained on SAR data reduced the loss of resolution, which supports training both SR models on SAR images.

L3 [3] → **P1, EVAL-02, EVAL-03:** As in [3], we train the detector separately for each input condition and examine how the mAP results change depending on object size.

4. References

[1] S. Wei, X. Zeng, Q. Qu, M. Wang, H. Su, and J. Shi, "HRSID: A high-resolution SAR images dataset for ship detection and instance segmentation," *IEEE Access*, vol. 8, pp. 120234–120254, 2020, doi: 10.1109/ACCESS.2020.3005861.

[2] P. Addabbo *et al.*, "Super-resolution of synthetic aperture radar complex data by deep-learning," *IEEE Access*, vol. 11, pp. 23647–23658, 2023, doi: 10.1109/ACCESS.2023.3251565.

[3] J. Shermeyer and A. Van Etten, "The effects of super-resolution on object detection performance in satellite imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Long Beach, CA, USA, 2019, pp. 1432–1441, doi: 10.1109/CVPRW.2019.00184.

[4] T. Zhang *et al.*, "SAR ship detection dataset (SSDD): Official release and comprehensive data analysis," *Remote Sensing*, vol. 13, no. 18, Art. no. 3690, 2021, doi: 10.3390/rs13183690.

[5] Association for Computing Machinery, "ACM Code of Ethics and Professional Conduct," 2018. Accessed: [Oct. 23, 2026]. [Online]. Available: https://www.acm.org/code-of-ethics

[6] IEEE, "IEEE Code of Ethics," IEEE Policies, Sec. 7.8. Accessed: [Oct. 23, 2026]. [Online]. Available: https://www.ieee.org/about/corporate/governance/p7-8.html

[7] IEEE-CS/ACM Joint Task Force on Software Engineering Ethics and Professional Practices, "Software Engineering Code of Ethics and Professional Practice," 1999. Accessed: [Oct. 23, 2026]. [Online]. Available: https://www.computer.org/education/code-of-ethics